

Air Pollution Data Analysis

RStudio question sheet based on the supplied global air pollution dataset

Instructions

Use RStudio to write reproducible code for each question. Label all output tables and charts clearly. For Question 7, only compare countries with at least 20 city records.

Data preparation

1. Import the CSV file into R and inspect its rows, columns, column names, and data types.
2. Identify missing values in every column. Create a cleaned dataset by removing records with a missing Country or City.
3. After cleaning, how many country groups and city records remain?

Descriptive analysis

4. Calculate the mean and median of the overall AQI Value. What does the difference suggest about the distribution?
5. Count the records in each AQI Category. Which category occurs most often?
6. Identify the five countries with the highest number of city records.
7. For countries with at least 20 city records, identify the five countries with the highest average AQI Value.
8. Find the city record with the maximum AQI Value. Report country, city, AQI value, and AQI category.
9. Calculate the number and percentage of records classified as Moderate or worse.

Relationships and visualisation

10. Create a correlation matrix for overall AQI, PM2.5 AQI, Ozone AQI, NO2 AQI, and CO AQI. Which measure has the strongest relationship with overall AQI?
11. Create a scatter plot of PM2.5 AQI Value against overall AQI Value. Add a linear trend line and describe the relationship.
12. Create a horizontal bar chart showing the five countries with the highest average AQI from Question 7.